MATH 3042 Midterm test- VA- Solution

February 24, 2026 10:24 AM

MATH 3042- Midterm test Full Name:
Time: 120 minutes

1. A system administrator monitors response times (in milliseconds) for a web server over 145 requests. A numeric and visual summary data is as follows:

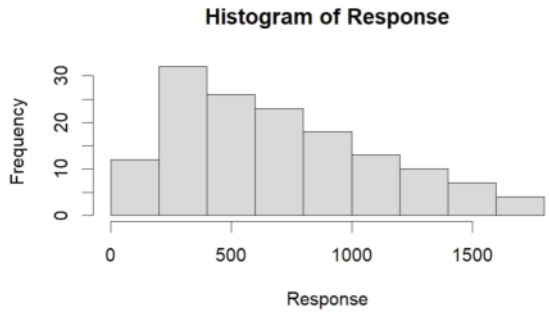
summary(Response)

Min	Q1	Med	Mean	Q3	Max
120.0	338.0	622.0	695.9	968.0	1745.0

length(Response): 145

sd(Response): 417.1723

hist(Response)



Based on the information above:

a. (3 marks) By calculating the upper and lower fence[1] determine if there are outliers among the Response data.

$IQR = 968 - 338 = 630$
$UF = 968 + 1.5 \times 630 = 1913$
$LF = 338 - 1.5 \times 630 = -607$
There are no outliers. Min and Max are within the fences.

b. (2 mark) Determine the z-score of a response time of 535 ms

$$z = \frac{535 - 695.9}{417.17} = -0.386$$

c. (1 mark) Does the distribution appear to be symmetric, right skewed, or left skewed?

Right Skewed (both from histogram and from the fact that Mean> Median)

d. (1 mark) The last column on the histogram ($1600 \leq Response \leq 1800$) has a frequency of 4. What is the percentage of response times in that column?

$$percentage = \frac{4}{145} = 2.76\%$$

- [1] Lower fence = Q1 - 1.5×IQR
- Upper fence = Q3 + 1.5×IQR
- IQR = Q3-Q1
- $z = \frac{x - \bar{x}}{s}$

2. (4 marks) A CI/CD pipeline can fail at build stage (B), test stage (T), or deployment stage (D):

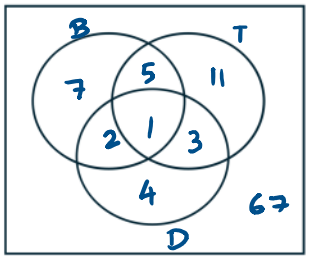
Given that: $P(B) = 0.15$, $P(T) = 0.20$, $P(D) = 0.10$, $\boldsymbol{P(B \mid T) = 0.3}$,

$$P(B \cap D) = 0.03, P(T \cap D) = 0.04, \quad P(B \cap T \cap D) = 0.01$$

Determine the P(pipeline succeeds).

Note: Instead of $P(B \cap T)$ you have been given the $P(B|T)$. Use it to find $P(B \cap T)$

$$P(B|T) = \frac{P(B \cap T)}{P(T)}, \qquad P(B \cap T) = (0.20)(0.3) = 0.06$$



$P(\bar{B} \cap \bar{D} \cap \bar{T}) = P(\overline{B \cup D \cup T}) = 67\%$

3. (3 marks) 26 key values are randomly assigned into 200 hash buckets. What is the probability of at least one hash collision (two key values assigned to the same bucket)?

$$P(at\ least\ one\ hash\ collision) = 1 - P(no\ collisions) = 1 - \frac{P(200,26)}{200^{26}} = 1 - 0.1829 = 0.817$$
$$\approx 81.71\%$$

4. (8 marks) A cloud storage system performs nightly backups of 80 databases. Each backup has a 97% probability of completing successfully without errors.

a) What is the probability that exactly 75 backups succeed?

$$P(x = 75) = dbinom(75, 80, 0.97) = 0.059$$

b) What is the probability that at least 78 backups succeed?

$$P(x \geq 78) = 1 - P(x \leq 77) = 1 - 0.43188 = 0.56812$$

c) Calculate the expected number of successful backups and the standard deviation. Calculate each. $\mu = np$, $\sigma^2 = npq$

$$\mu = (80)(0.97) = 77.6, \qquad \sigma = \sqrt{(80 \times 0.97 \times 0.03)} = \sqrt{2.328} = 1.5258$$

d) Identify all the unusually low number of successful backups. Use the $Z < -2$ criteria

$-2 = \frac{x-77.6}{1.5258}$, $x = 74.55$ all successful back up counts less than or equal to 74.55 are unusually low

Please use as needed:

```
> pbinom(75, 80, 0.97)
[1] 0.09279325
> dbinom(75, 80,0.97)
[1] 0.05948687
> pbinom(78, 80, 0.97)
[1] 0.6961936
> pbinom(77, 80, 0.97)
[1] 0.4318767
```

5. (3 marks) A REST API has 40 endpoints, of which 14 contain security vulnerabilities. A penetration tester randomly selects 8 endpoints to examine. Use a calculator and determine the probability that at least 1 vulnerable endpoint is selected? Please write down what you are calculating.

$$P(x \geq 1) = 1 - P(x = 0) = 1 - \frac{C(14,0) \times C(26,8)}{C(40,8)} = 1 - 0.0203 = 0.9797$$

6. (4 marks) For each scenario, identify which probability distribution is most appropriate (Binomial, Hypergeometric, Geometric, or Poisson) and explain why. No need to solve the problem, just identify the appropriate distribution.

a. A web scraper sends requests to a server and encounters a rate-limit error on average 4 times per hour. You want to find the probability of exactly 6 rate-limit errors in the next hour.

Poisson

b. A database table has 500 rows, 80 of which contain duplicate entries. A data analyst randomly pulls 30 rows for a spot-check audit and wants to know the probability that exactly 7 of them are duplicates.

Hypergeometric

c. You are stress-testing a network connection that has a 15% chance of dropping a packet on any given transmission. You want to know the probability that the first dropped packet occurs on exactly the 4th transmission.

Geometric

d. A machine learning pipeline runs 50 independent preprocessing steps, each with a 98% chance of completing without error. You want the probability that exactly 48 steps complete successfully.

Binomial

7. The time (in minutes) that a software build takes to complete on a CI/CD pipeline is *exponentially* [2]distributed, with a mean of 8 minutes.

A. (2 marks) What is the probability that a build completes in under 5 minutes?

$$P(x < 5) = \left(1 - e^{-\frac{5}{8}}\right) = 0.4647$$

B. (2 marks) What is the probability that a build takes between 6 and 12 minutes?

$$P(6 < x < 12) = \left(1 - e^{-\frac{12}{8}}\right) - \left(1 - e^{-\frac{6}{8}}\right) = 0.2492$$

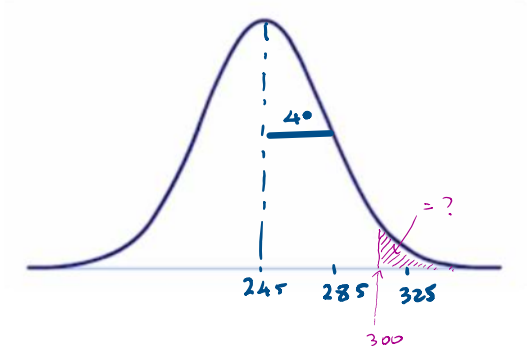
C. (2 marks) What is the 80th percentile of build completion times?

$$1 - e^{-\frac{t}{8}} = 0.80\,, \qquad t = -8\ln 0.2 = 12.9\ minutes$$

[2] $cdf(t) = 1 - e^{-\frac{t}{\beta}}$

8. The response time (in milliseconds) of a web API under normal load is normally distributed, with a mean of 245 ms and a standard deviation of 40 ms.

a. (3 marks) What is the probability that a randomly-chosen API call has a response time greater than 300 ms? Please place the relevant numbers on the given graph and shade the region that corresponds to the probability in question. Write the filled in R command(s) that would solve this problem.



$P(x > 300) = 1 - pnorm(300, 245, 40)$

b. (3 marks) What values mark the middle 80% of API response times? Given:

```
> qnorm(0.10)
[1] -1.281552
> qnorm(0.20)
[1] -0.8416212
```

$\pm 1.281552 = \frac{x - 245}{40}$, $x_{low} = 193.74\ ms$, $x_{high} = 296.26\ ms$

9. In a city, historically 42% of voters support Candidate A. A polling organization conducts multiple polls.

a. (2 marks) Calculate the mean and standard deviation of the sampling distribution of proportions.

$$\mu_{\hat{p}} = p \, , SE = \sqrt{\left[\frac{p(1-p)}{n}\right]}$$

Is $(200) \times 0.42 \geq 10? \, Yes$
Is $(200) \times (1 - 0.42) \geq 10? \, Yes$
We can used CLT: $\mu_{\hat{p}} = 0.42 \;\; , SE = 0.034899$

b. (2 marks) Using the normal approximation, calculate the probability that a random sample of 200 voters would show more than 38% support for Candidate A. Please write filled-in R command(s) for this part.

$$P(\hat{p} > 0.38) = 1 - pnorm(0.38, 0.42, 0.034899)$$

c. (2 marks) At a 5% significance level, is it unusual in an election to observe a 62% support for candidate A? Please use qnorm(0.05) = -1.644854 in your reasoning

Let's calculate the z-score of the observed p-hat:

$$Z = \frac{0.62 - 0.42}{0.034899} = 5.73$$

This is highly unusual. It is many standard deviations (5.7) above the center of the distribution.